

# Subquery

## Subquery – one row

|   | first_name | last_name | department_id |
|---|------------|-----------|---------------|
| 1 | Matthew    | Weiss     | 50            |
| 2 | Adam       | Fripp     | 50            |
| 3 | Payam      | Kaufling  | 50            |
| 4 | Shanta     | Vollman   | 50            |
| 5 | Kevin      | Mourgos   | 50            |
| 6 | Julia      | Nayer     | 50            |

```
select
    first_name,
    last_name,
    department_id
from employees
where
    department_id =
        (select department_id from departments where department_name = 'Shipping')
```

## Subquery – one row - error

Results Messages

Msg 512, Level 16, State 1, Line 1

Subquery returned more than 1 value. This is not permitted when the subquery follows =, !=, <, <=, >, >= or when the subquery is used as an expression.

```
select
    first_name,
    last_name,
    department_id
from employees
where
    department_id =
    (select department_id from departments
     where department_name in ('Shipping', 'Administration'))
```



## Subquery – many in one query

|   | first_name | last_name | department_id |
|---|------------|-----------|---------------|
| 1 | David      | Williams  | 60            |
| 2 | Valli      | Jackson   | 60            |
| 3 | Diana      | Nguyen    | 60            |
| 4 | Nandita    | Sarchand  | 50            |
| 5 | Alexis     | Bull      | 50            |
| 6 | Sarah      | Bell      | 50            |
| 7 | Jennifer   | Whalen    | 10            |

```
select
    first_name,
    last_name,
    department_id
from employees
where salary between (select min_salary from jobs
                      where job_title = 'Programmer')
and (select max_salary from jobs
     where job_title = 'Stock Clerk')
```

## Subquery – in having

|   | department_id | (No column name) |
|---|---------------|------------------|
| 1 | NULL          | 7000.00          |
| 2 | 10            | 4400.00          |
| 3 | 40            | 6500.00          |
| 4 | 70            | 10000.00         |

```
select
    department_id,
    sum(salary)
from employees
group by department_id
having sum(salary) < (select sum(salary) from employees
                      where department_id = 20)
```

## Subquery – operator ANY

|   | first_name | last_name | salary   | department_id |
|---|------------|-----------|----------|---------------|
| 1 | Steven     | King      | 24000.00 | 90            |
| 2 | Neena      | Yang      | 17000.00 | 90            |
| 3 | Lex        | Garcia    | 17000.00 | 90            |
| 4 | Alexander  | James     | 9000.00  | 60            |
| 5 | Nancy      | Gruenberg | 12008.00 | 100           |
| 6 | Daniel     | Faviet    | 9000.00  | 100           |
| 7 | John       | Chen      | 8200.00  | 100           |
| 8 | Ismael     | Sciarra   | 7700.00  | 100           |

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees
where salary >ANY (select salary from employees
                    where department_id = 20)
```

## Subquery – operator ALL

|   | first_name | last_name | salary   | department_id |
|---|------------|-----------|----------|---------------|
| 1 | Steven     | King      | 24000.00 | 90            |
| 2 | Neena      | Yang      | 17000.00 | 90            |
| 3 | Lex        | Garcia    | 17000.00 | 90            |
| 4 | John       | Singh     | 14000.00 | 80            |
| 5 | Karen      | Partners  | 13500.00 | 80            |

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees
where salary >ALL (select salary from employees
                    where department_id = 20)
```

## Subquery – correlation

|   | first_name | last_name | salary   | department_id |
|---|------------|-----------|----------|---------------|
| 1 | Michael    | Martinez  | 13000.00 | 20            |
| 2 | Den        | Li        | 11000.00 | 30            |
| 3 | Nandita    | Sarchand  | 4200.00  | 50            |
| 4 | Alexis     | Bull      | 4100.00  | 50            |
| 5 | Kelly      | Chung     | 3800.00  | 50            |
| 6 | Jennifer   | Dilly     | 3600.00  | 50            |
| 7 | Sarah      | Bell      | 4000.00  | 50            |

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees e
where salary > (select avg(salary) from employees
                where department_id = e.department_id)
```

## Subquery – correlation - EXISTS

|   | first_name | last_name   | salary  | department_id |
|---|------------|-------------|---------|---------------|
| 1 | Matthew    | Weiss       | 8000.00 | 50            |
| 2 | Adam       | Fripp       | 8200.00 | 50            |
| 3 | Payam      | Kaufling    | 7900.00 | 50            |
| 4 | Shanta     | Vollman     | 6500.00 | 50            |
| 5 | Kevin      | Mourgos     | 5800.00 | 50            |
| 6 | Julia      | Nayer       | 3200.00 | 50            |
| 7 | Irene      | Mikkilineni | 2700.00 | 50            |
| 8 | James      | Landry      | 2400.00 | 50            |

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees e
where EXISTS (select * from departments
where department_name = 'Shipping' and department_id = e.department_id)
```

## Subquery – correlation – NOT EXISTS

|   | first_name | last_name | salary   | department_id |
|---|------------|-----------|----------|---------------|
| 1 | Steven     | King      | 24000.00 | 90            |
| 2 | Neena      | Yang      | 17000.00 | 90            |
| 3 | Lex        | Garcia    | 17000.00 | 90            |
| 4 | Alexander  | James     | 9000.00  | 60            |
| 5 | Bruce      | Miller    | 6000.00  | 60            |
| 6 | David      | Williams  | 4800.00  | 60            |

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees e
where EXISTS (select * from departments
where department_name = 'Shipping' and department_id = e.department_id)
```

## Problem

### Messages

Msg 207, Level 16, State 1, Line 6  
Invalid column name 'full\_salary'.

```
select
    first_name,
    last_name,
    salary + (salary * isnull(commission_pct,0)) full_salary
from employees e
where full_salary > 15000
```

Problem –  
resolve by  
subquery

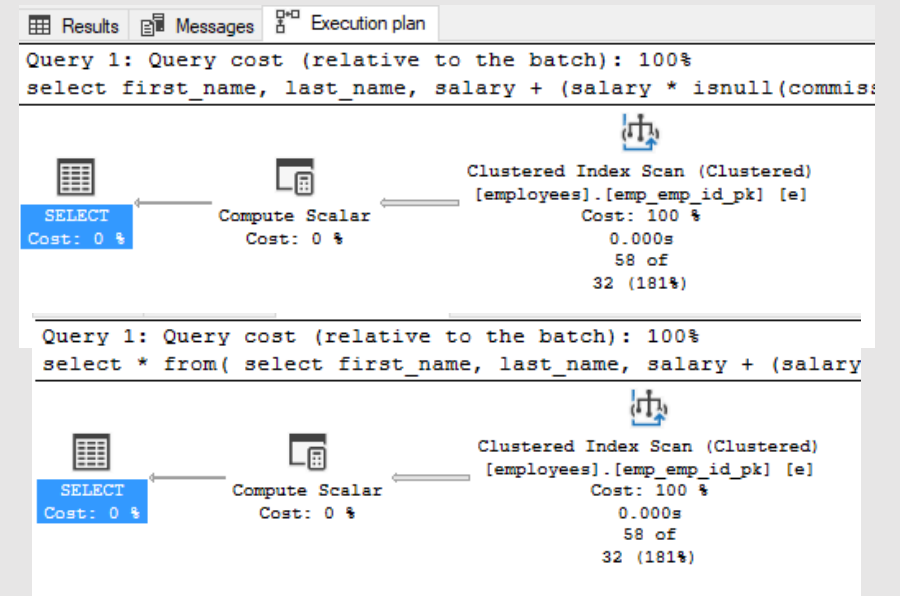
|   | first_name | last_name | full_salary |
|---|------------|-----------|-------------|
| 1 | Steven     | King      | 24000.0000  |
| 2 | Neena      | Yang      | 17000.0000  |
| 3 | Lex        | Garcia    | 17000.0000  |
| 4 | John       | Singh     | 19600.0000  |
| 5 | Karen      | Partners  | 17550.0000  |
| 6 | Alberto    | Errazuriz | 15600.0000  |

```
select * from(  
  select  
    first_name,  
    last_name,  
    salary + (salary * isnull(commission_pct,0)) full_salary  
  from employees e ) t1  
where t1.full_salary > 15000
```

## Compare execution plans

```
select
    first_name,
    last_name,
    salary + (salary * isnull(commission_pct,0))
full_salary
from employees e
where salary + (salary * isnull(commission_pct,0)) > 15000
```

```
select * from(
    select
        first_name,
        last_name,
        salary + (salary * isnull(commission_pct,0)) full_salary
    from employees e ) t1
where t1.full_salary > 15000
```



## Subquery in select

|   | first_name | last_name | dep_name  |
|---|------------|-----------|-----------|
| 1 | Steven     | King      | Executive |
| 2 | Neena      | Yang      | Executive |
| 3 | Lex        | Garcia    | Executive |
| 4 | Alexander  | James     | IT        |
| 5 | Bruce      | Miller    | IT        |
| 6 | David      | Williams  | IT        |
| 7 | Valli      | Jackson   | IT        |
| 8 | Diana      | Nguyen    | IT        |
| 9 | Nancy      | Gruenberg | Finance   |

```
select
    first_name,
    last_name,
    (select department_name from departments where department_id =
        e.department_id) dep_name
from employees e
```

## Subquery in from

|   | first_name | last_name |
|---|------------|-----------|
| 1 | Jennifer   | Whalen    |

```
select
  t1.first_name,
  t1.last_name
from (select first_name, last_name, department_id
      from employees
      where department_id =10) t1
```